Mohammad Shayan

Telecommunications Engineering

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SUMMARY

Telecommunications Engineering student with hands-on IoT and embedded systems experience. Award-winning developer (Global IoT Winner, International IoT 2nd Runner-Up) specializing in sensor integration, real-time systems, and edge computing. Proven track record delivering production systems processing 500+ users with 99%+ uptime.

EDUCATION

Sep 2022 – Aug 2026

NED University of Engineering & Technology, Karachi
Bachelor of Engineering — Telecommunications Engineering

Sep 2022 – Sep 2025

Aptech Pakistan, Karachi
Advance Diploma in Software Engineering 72%

Aug 2020 – Aug 2022

Habib Public School, Karachi
Intermediate in Computer Science A Grade

EXPERIENCE

Dec 2025 – Jan 2026

Azfam Technologies, Karachi
IoT Web Development Intern

- Automated manual climate control processes by engineering an IoT-based AC system, reducing on-site interventions by 60% and enabling 24/7 remote access.
- Designed a real-time web dashboard with Firebase integration, achieving sub-second latency for sensor data updates across 10+ connected devices.
- Deployed responsive UI for cross-device compatibility, improving operator efficiency and reducing average response time from 15 min to under 2 min.

SKILLS

Technical: Embedded Systems, IoT, Wireless Communications, RTOS, LoRa, ESP32, Raspberry Pi

Programming: Python, C/C++, JavaScript, PHP, HTML & CSS

Frameworks: Flask, OpenCV, Firebase, MySQL, Bootstrap, Git, Linux

Soft Skills: Leadership, Teamwork, Problem-Solving

PROJECTS

SHARC — Self-Healing Anti-Jamming Radio Communication (Final Year Project, In Progress)
ESP32-S3 · LoRa SX1276/SX1278 · Edge ML · Imitation Learning

- Addressing unreliable long-range communication in interference-prone environments by developing an AI-driven anti-jamming system targeting 95%+ jamming detection accuracy.
- Implementing edge-based ML with imitation learning to enable real-time adaptive channel switching in under 50ms, maintaining link stability under active jamming.

https://nexsys.live

DoorCam — Smart IoT Security System

Python · OpenCV · Flask · Firebase · Twilio · Raspberry Pi 4B

- Solved unauthorized access detection by building a facial recognition system achieving 97% accuracy across 50+ enrolled users with sub-2s identification time.
- Eliminated security blind spots by integrating PIR, ultrasonic, and LDR sensors, reducing false positive alerts by 80% through multi-sensor fusion logic.
- Enabled instant incident response via Twilio SMS/push alerts within 3 seconds; **2nd Runner-Up**, International IoT Ignite 2025 (APTECH).

https://github.com/TechWizardsTechwiz6/Wavebell

WeatherNinja — Smart Weather Monitoring Station

Python · Firebase · Raspberry Pi 4B · DHT11 · BMP280 · YL-83

- Replaced manual weather logging by deploying 5 sensors monitoring temperature, humidity, pressure, rain, and light with 99.9% data capture uptime.
- Delivered real-time alerts for extreme conditions within 5 seconds via SMS, enabling proactive response to weather anomalies.
- **Winner**, Global IoT Solution Development (APTECH 2024)

https://github.com/MohammadShayan1/techwiz

NED Directorate of Industrial Liaison — Web Platform

HTML · CSS · JavaScript · PHP · MySQL

- Streamlined university-industry connections by building a full-stack platform that processed 500+ internship applications, reducing manual admin workload by 70%.
- Implemented SEO optimization and FYDP management modules, increasing platform discoverability and supporting 200+ final year project submissions.

https://github.com/MohammadShayan1/NEDDIL-main

TE Links Website — Telecommunication Student Society (Manager Technical)

HTML · Bootstrap · JavaScript · PHP · MySQL · AOS

- Led technical development of a 7-page responsive society website, increasing online engagement by 3x and serving 100+ active student members.

https://telinks.org

NEDMUN-VI — Conference Registration System

HTML · Bootstrap · PHP · MySQL

- Eliminated paper-based registration by building a digital platform with dual delegate flows, processing 300+ registrations with zero data loss.
- Enabled real-time admin analytics and CSV export, reducing registration management overhead by 50% compared to manual processes.

https://nedmun.telinks.org

AWARDS & HONORS

Nov 2025

2nd Runner-Up — International IoT Ignite | APTECH

DoorCam: IoT security system with 97% facial recognition accuracy, multi-sensor fusion, and real-time alerts — competing against 80+ international teams.

Nov 2024

Winner — Global IoT Solution Development | APTECH

WeatherNinja: Smart weather station for real-time monitoring, 99.9% uptime, and instant SMS alert system.